

Effect of metformin compared with hypocaloric diet on serum C-reactive protein level and insulin resistance in obese and overweight women with poly...

AIM: The aim of the present study was to investigate the efficacy of Metformin compared with a hypocaloric diet on C-reactive protein (CRP) level and markers of insulin resistance in obese and overweight women with polycystic ovary syndrome (PCOS). MATERIAL AND METHODS: Forty women with body mass index ≥ 27 and PCOS were randomly allocated to receive either Metformin or hypocaloric diet and were assessed before and after a treatment period of 12 weeks. High-sensitivity CRP (hs-CRP) and markers of insulin resistance (IR), homeostasis model assessment-IR, quantitative insulin-sensitivity check index and fasting glucose to insulin ratio were evaluated in each patient. RESULTS: A total of 10 subjects did not complete the trial (three patients in the Metformin group and seven patients in the diet group) and a total of 30 subjects completed the trial (17 subjects in the Metformin group and 13 subjects in the diet group). Serum concentration of hs-CRP significantly decreased in both the Metformin (5.29 ± 2.50 vs 3.81 ± 1.99 , $P = 0.008$) and diet groups (6.08 ± 2.14 vs 4.27 ± 1.60 , $P = 0.004$). There were no significant differences in mean hs-CRP decrement between the two groups. Decrease in hs-CRP levels was significantly correlated with waist circumference in the diet group ($r = 0.8$, $P < 0.001$). The effect of a hypocaloric diet with 5-10% weight reduction on markers of insulin resistance (homeostasis model assessment-IR, fasting glucose to insulin ratio, quantitative insulin-sensitivity check index) was better than Metformin therapy ($P = 0.001$). CONCLUSIONS: Although weight reduction has equal efficacy with Metformin in decreasing serum hs-CRP levels, it was significantly more effective in improving insulin resistance in obese and overweight PCOS women. © 2012 The Authors. Journal of Obstetrics and Gynaecology Research © 2012 Japan Society of Obstetrics and Gynecology.